

Sarvam AI TTS Dataset Report

1. Project Overview

I built a complete Text-to-Speech (TTS) training dataset for Indian English (en-IN) and Hindi (hi-IN) using YouTube-sourced clean, single-speaker educational and lecture audio clips. The pipeline includes automatic audio downloading, audio format conversion, silence-based segmentation, transcription via Sarvam ASR, emotion/style tagging using Sarvam LLM, automated quality filtering, manual quality control, and packaging as a HuggingFace-compatible dataset.

2. Dataset Composition

The final processed and validated dataset has the following statistics:

* **Accept

Note: Although the original target was approximately 60 minutes, I prioritized validated clean clips and strict audio quality over blindly padding the dataset with noisy or poorly segmented speech to meet the target duration under deadline constraints.

3. Pipeline Overview

The end-to-end dataset generation pipeline consists of the following stages:

1. YouTube

4. Source Selection Criteria

To ensure high-quality speech data, sources were selected based on the following rules:

* **Single

5. Audio Processing and Segmentation

Source audio was converted to mono WAV at 22050 Hz. Segmentation was performed using pydub's silence-based splitter (`min\_silence\_len=600ms`, `silence\_thresh=audio.dBFS-14`). Any segments with mid-word cuts, overlapping voices, or excessive ambient noise were identified and rejected during the QC phase.

6. Sarvam API Usage

* **Sarvam ASR**: Transcribed all segments using the `saarika:v2.5` model.

* **Sarvam
selected o
maximum v

7. Emotion/Style Tagging

Allowed tags were restricted to the following set: neutral, narrative, excited, angry, formal, happy, sad, calm, whisper.

The final di

8. Manual QC and Fast QC

We prioritized transcript correctness and clean audio near the deadline using an automated risk-filtering strategy:

* **High-
indicating n

9. Iterations and Improvements

* Fixed encoding issues during Hindi text transcription.

* Switc
conversatio

10. What Worked

* Sarvam ASR provided highly accurate first-pass transcriptions for both English and Hindi.

* Silence-

11. What Did Not Work

* YouTube rate limits and bot-detection schemes temporarily blocked yt-dlp.

* Some H
correctly.

12. Quality Decisions

* Rejected clips containing mid-word cuts, background noise, and overlapping audio.

* Retained

13. Final Dataset Links

* **HuggingFace Dataset**:
https://huggingface.co/datasets/champTUSHARg007/indian-tts-dataset

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14. Limitations

* Total duration (28.79 minutes) is lower than the 60-minute target.

* Hindi clip

15. Future Improvements

* Add 3-4 more clean, single-speaker Hindi lecture sources to balance the language representation.

* Run full